



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KAVI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

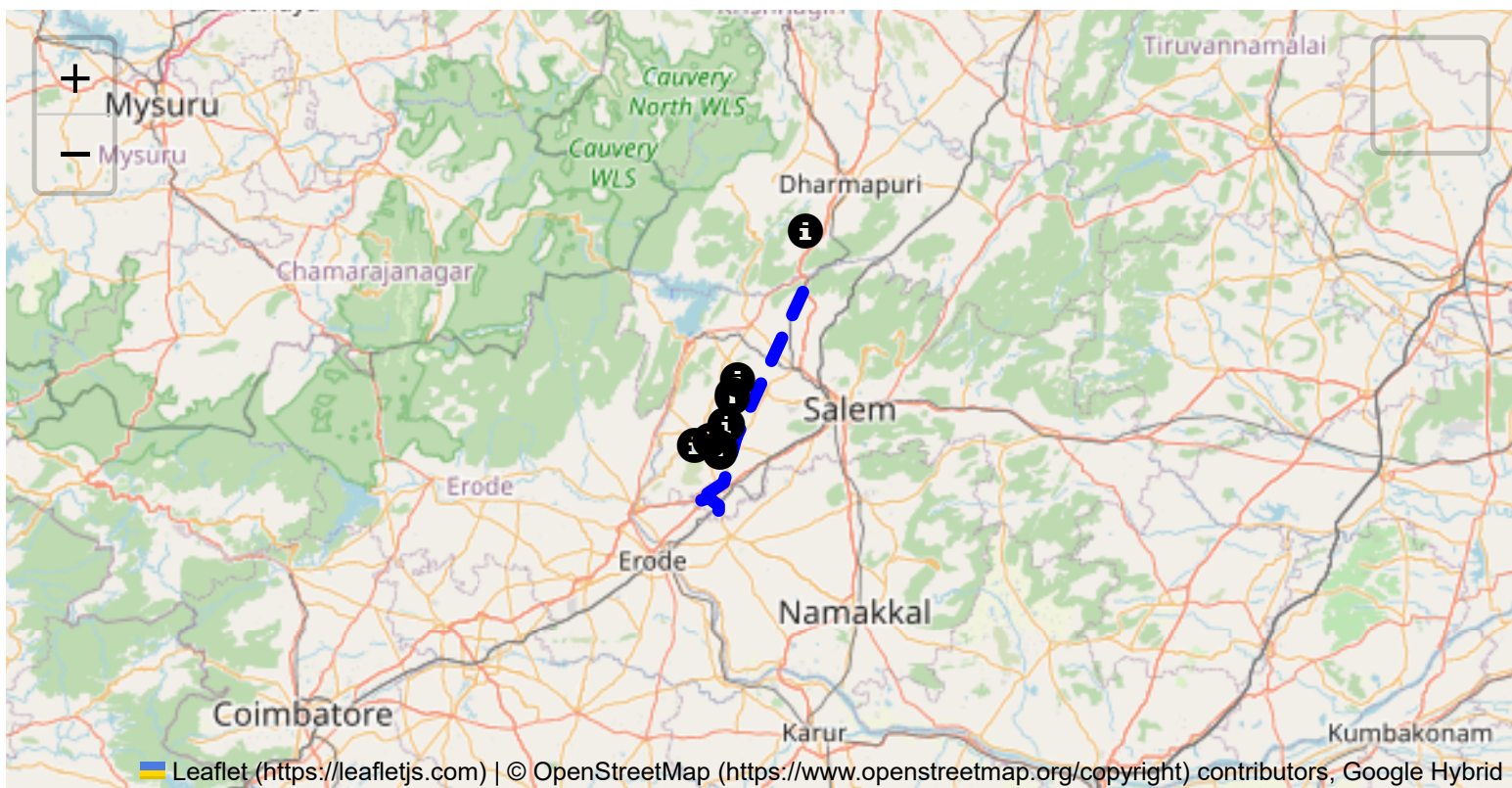
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

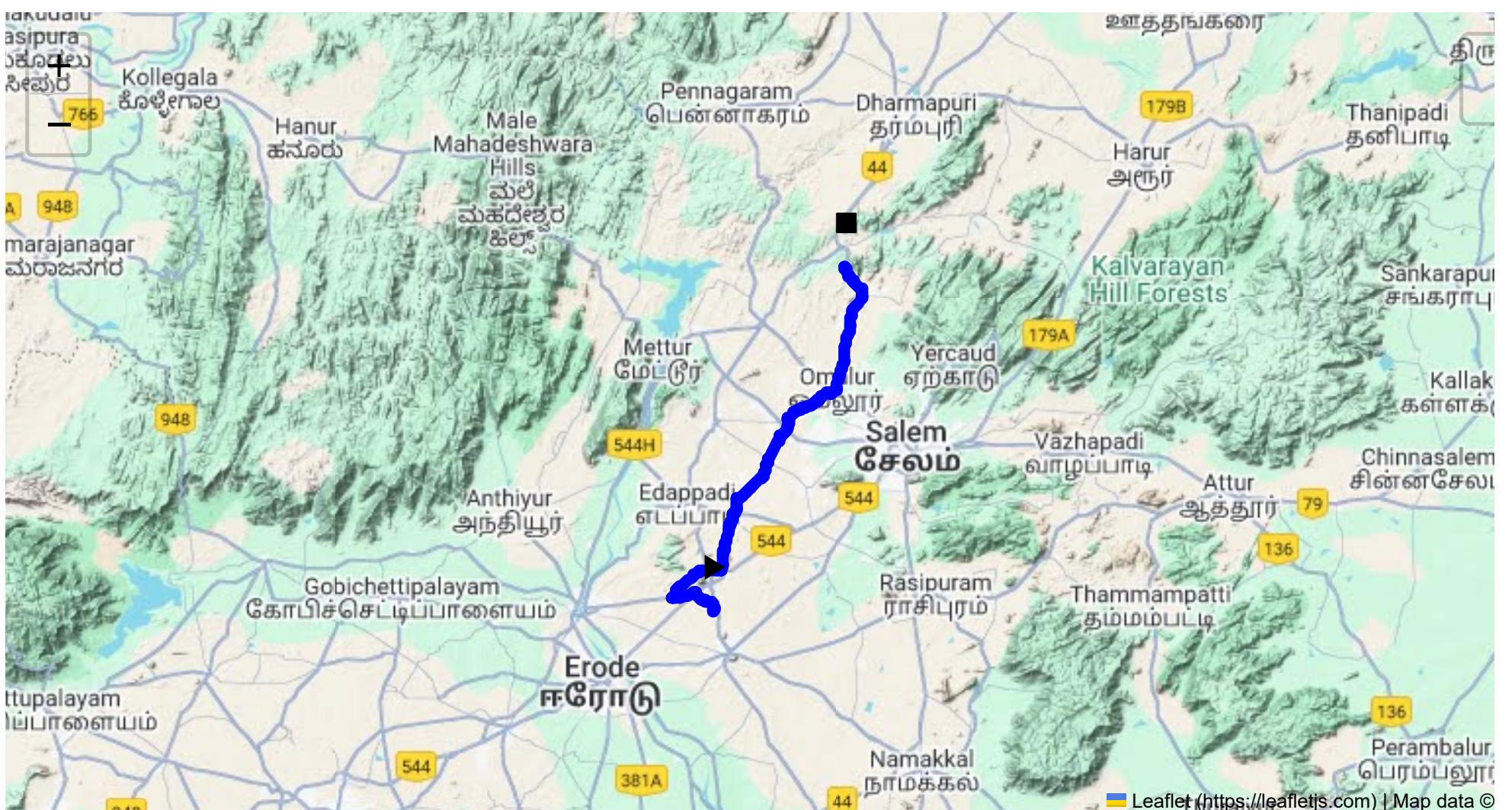
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 74.79 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours
Start: (11.4381, 77.8734)
End: (11.910435, 78.057648)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route extends approximately 74.79 kilometers starting from Sangagiri in Tamil Nadu and ending in Jodukuli, Tamil Nadu. The journey passes through several key locations including Sankari RS, Sankari West, Padaiveedu, Devannagoundanur, Ayveli, Erumaipatti, and Koranampatti, primarily along SH 86. This route mainly traverses rural and semi-urban areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. The route typically sees hot conditions most of the year with temperatures averaging between 25°C to 37°C. The monsoon period from June to September can present heavy rains which may lead to slippery roads and localized flooding, especially in low-lying areas. Infrequent cyclonic activity during the November to December period could exacerbate these conditions.

3. Analysis of Traffic Patterns

- **Peak Hours:** Commuter traffic is generally heavier between 8:00 AM - 10:00 AM and 5:00 PM - 7:00 PM.
- **Congestion-Prone Areas:** Expect congestion near urban centers like Sankari RS and at market areas in Sankari West and Erumaipatti due to local trade activities.

4. Assessment of Road Quality and Infrastructure

SH 86, the primary artery for this route, is generally in good condition but may suffer from potholes, especially post-monsoon. Some sections, particularly near smaller villages and junctions, may lack proper signage and light, posing a risk during nighttime driving.

5. Suggestions for Alternative Routes for Emergencies

In case of road closures or severe traffic, consider detours through smaller village roads around the main towns. However, these roads might be narrower and less maintained. The National Highway 47 is an available alternative on the eastern side should a larger diversion be necessary.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Indian regulations mandate specific routes for hazardous materials, often avoiding densely populated areas. Permits are required for transportation, ensuring compliance with local safety standards. Check for the latest updates on permissible routes and legal formalities.

7. Overview of Historical Incidents

The region has seen occasional incidents involving heavy vehicles, mostly attributable to human error or poor road conditions. There are fewer records of incidents specifically involving hazardous materials, indicating generally safe practices.

8. Environmental Considerations and Sensitive Areas

Pass through agricultural lands which may limit handling of hazardous materials due to crop contamination risks. Carefully navigate near water bodies to prevent environmental damage.

9. Analysis of Communication Coverage

- **Potential Dead Zones:** Rural areas between Ayveli and Devannagoundanur may have weak network signals. Prepare for communication loss, especially in remote stretches.

10. Estimated Emergency Response Times

Response times can vary:

- **Urban Areas (e.g., Sankari):** 20-30 minutes
- **Rural Areas:** 45 minutes to 1 hour

Authorities like local police, fire departments, and hospitals are relatively accessible in urban centers but response can be slower in rural sections.

12. Overall Summary of Risk Assessment

This route, while generally safe, presents typical hazards related to rural driving in India such as unpredictable traffic, potentially poor road conditions, and weather-related issues during the monsoon. Observance of local traffic rules, preparation for potential communication loss, and ready access to emergency contacts and alternative routes are key measures for ensuring safety. Proper maintenance of the vehicle and compliance with hazardous material transportation regulations will further mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45634, 77.81673	15 KM/Hr
11	Turn	Medium	11.49416, 77.88011	30 KM/Hr
12	Turn	Medium	11.49421, 77.88021	30 KM/Hr
13	Turn	High	11.49383, 77.88526	15 KM/Hr
14	Turn	High	11.49395, 77.88537	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.49409, 77.88536	15 KM/Hr
16	Turn	Medium	11.49435, 77.88499	30 KM/Hr
17	Turn	Medium	11.49465, 77.88488	30 KM/Hr
18	Turn	Medium	11.56096, 77.89931	30 KM/Hr
19	Turn	High	11.57035, 77.90223	15 KM/Hr
20	Turn	High	11.57041, 77.90221	15 KM/Hr
21	Turn	Medium	11.58279, 77.90658	30 KM/Hr
22	Turn	Medium	11.63833, 77.94818	30 KM/Hr
23	Turn	Medium	11.64171, 77.95184	30 KM/Hr
24	Turn	Medium	11.64243, 77.95198	30 KM/Hr
25	Turn	Medium	11.64721, 77.95118	30 KM/Hr
26	Turn	Medium	11.67858, 77.96834	30 KM/Hr
27	Turn	Medium	11.68915, 77.97734	30 KM/Hr
28	Turn	High	11.70227, 77.97864	15 KM/Hr
29	Turn	Medium	11.71921, 78.01133	30 KM/Hr
30	Turn	High	11.73651, 78.03878	15 KM/Hr
31	Turn	High	11.91102, 78.05871	15 KM/Hr
32	Turn	Medium	11.91093, 78.05828	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49416, 77.88011



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49421, 77.88021



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88526



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88536



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49435, 77.88499



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49465, 77.88488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57041, 77.90221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58279, 77.90658



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68915, 77.97734



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91102, 78.05871



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91093, 78.05828
